



STYLESENSE P

AI-Powered Fashion Recommendation System | Computer Vision
+ RAG + LLM

PROBLEM & SOLUTION



Users struggle to get context-aware outfit guidance from images alone. A photo captures appearance but lacks the semantic understanding needed to answer nuanced fashion questions — what works, what clashes, and what fits the occasion.



StyleSense bridges the gap by applying YOLO object detection to identify individual clothing items in an uploaded outfit image, transforming visual data into structured, meaningful outfit context.



The detected outfit context is passed through a RAG pipeline that retrieves relevant fashion knowledge from ChromaDB, enabling the LLM to generate a grounded, accurate Arabic fashion recommendation tailored to the user's query.



SYSTEM ARCHITECTURE

Frontend & Backend

- Streamlit frontend for image upload and user interaction
- FastAPI backend handles all processing requests
- User uploads outfit image and submits a fashion question
- Clean REST API communication between layers



Detection & Retrieval

- YOLO model detects clothing items from uploaded image
- Structured outfit context is built from detected categories
- ChromaDB vector store retrieves top relevant knowledge chunks
- Semantic embeddings power similarity-based retrieval



Generation & Response

- LLM combines outfit context with retrieved knowledge
- Generates a grounded Arabic fashion recommendation
- Source references returned alongside the response
- Final recommendation delivered back to the user

AI PIPELINE



Stage 1 — Object Detection (YOLO): YOLO model detects up to 10 clothing categories from the uploaded outfit image, returning item labels and confidence scores to build a structured outfit context.



Stage 2 — Semantic Embedding & Retrieval (RAG): Sentence Transformers encode the outfit context into semantic embeddings. ChromaDB retrieves the top 3 most relevant fashion knowledge chunks using vector similarity search.

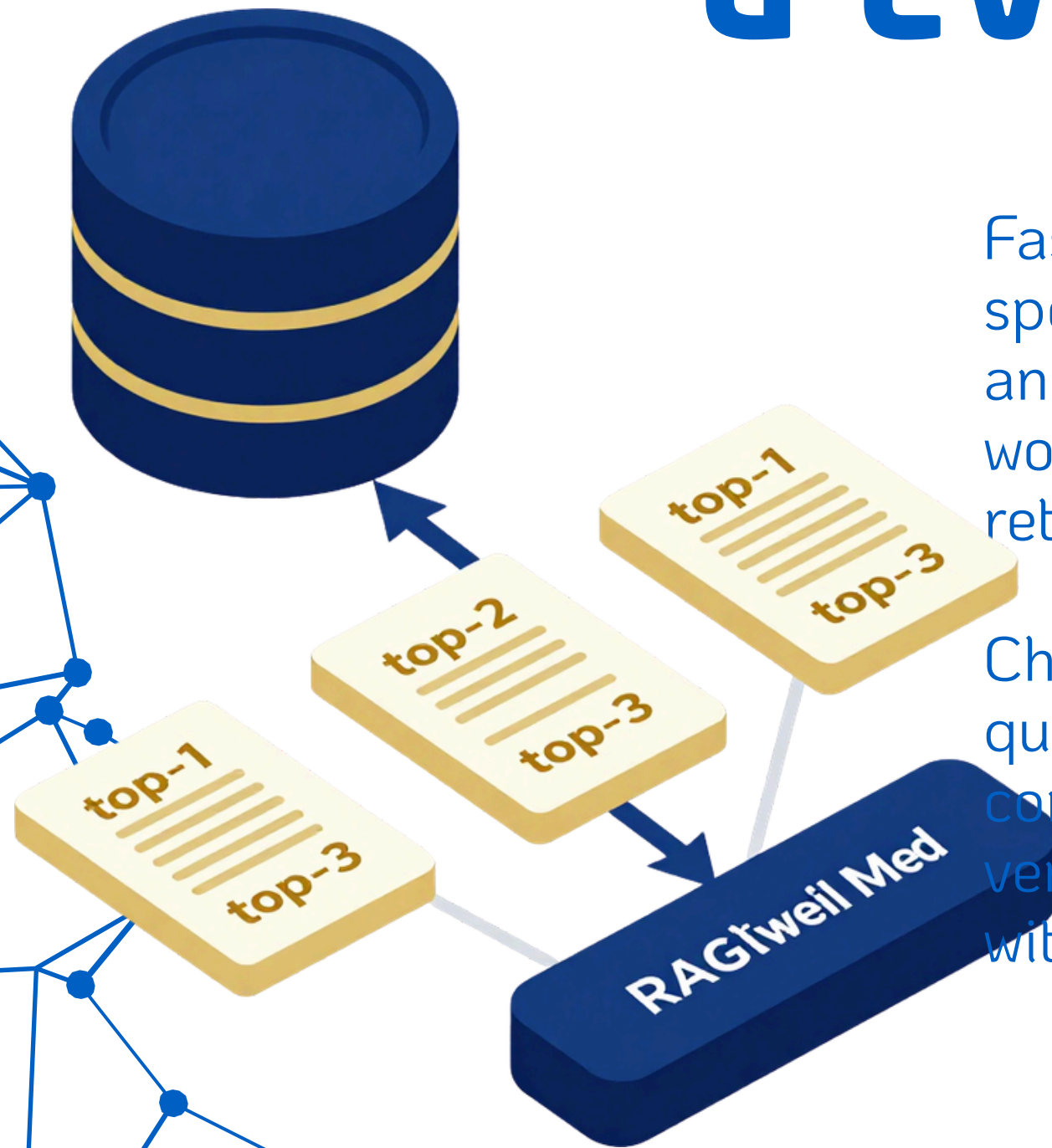


Stage 3 — LLM Generation: The large language model combines the detected outfit context with the retrieved knowledge chunks to generate a grounded, context-aware Arabic fashion recommendation.

RAG KNOWLEDGE BASE & EVALUATION

Fashion knowledge base covers 7 domains: casual, formal, sporty, color coordination, university style, interview attire, and accessories. Chunks are sized at 100 words with a 20-word overlap to preserve semantic continuity across retrieved segments.

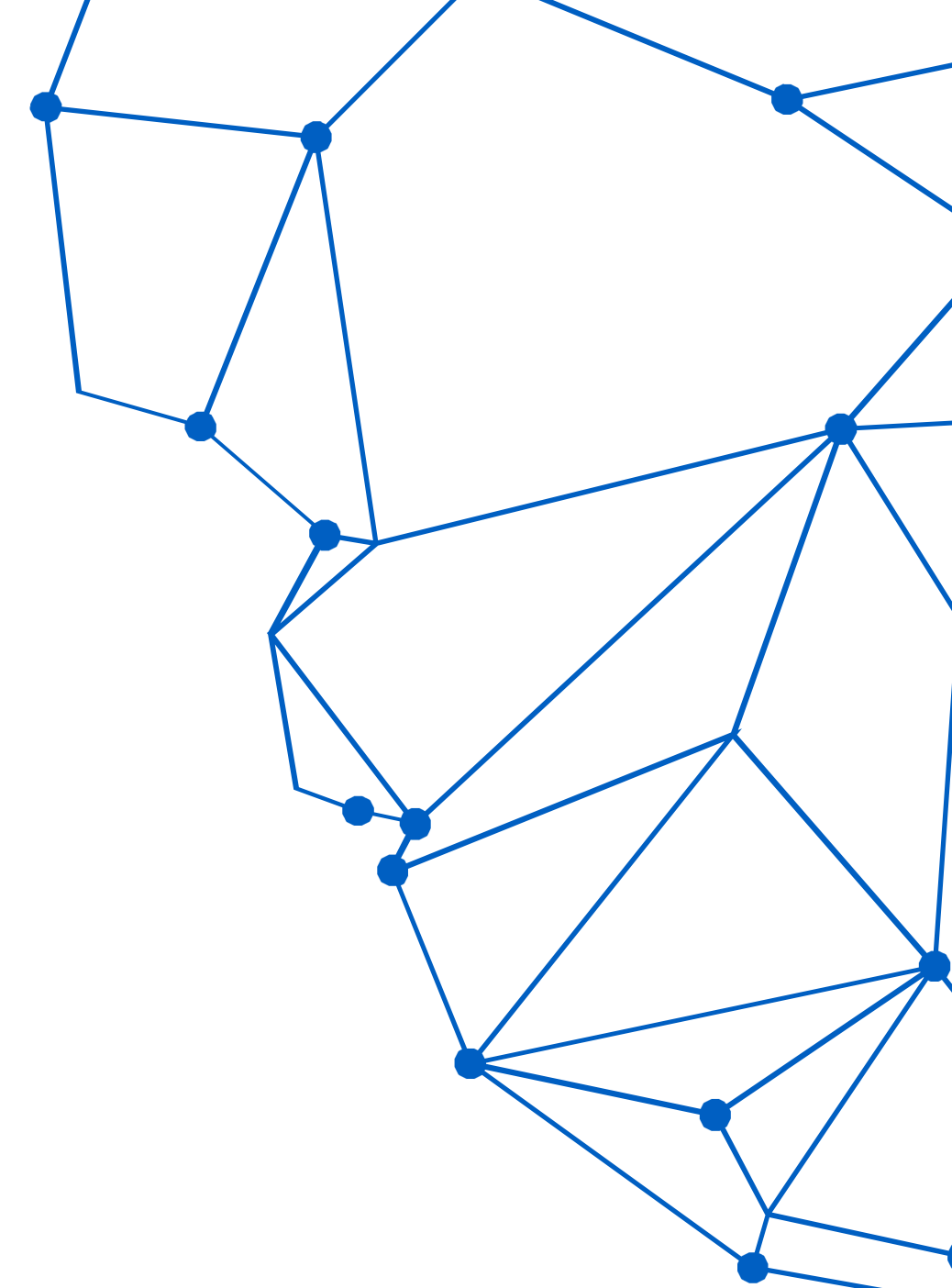
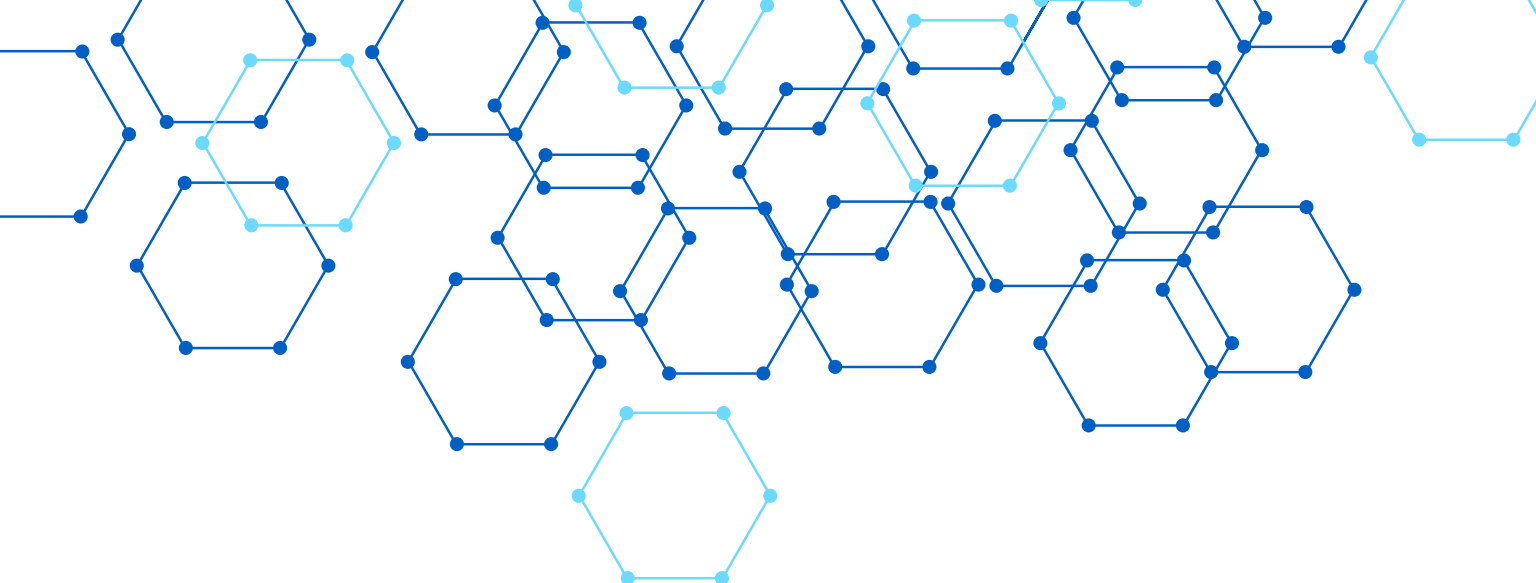
ChromaDB retrieves the top 3 most relevant chunks per query using semantic similarity. System evaluation was conducted on 10 representative fashion questions, verifying grounded and accurate Arabic recommendations with traceable source references.



APPLICATION DEMO

1. Upload Outfit Image
2. User uploads a photo of their outfit directly through the Streamlit interface.
3. YOLO Detection Results
4. The system detects clothing items with category labels and confidence scores (e.g., shirt 92%, jeans 88%).
5. Ask a Fashion Question
6. User types a context-aware question such as: "Is this outfit suitable for a university interview?"
7. Arabic Recommendation
8. StyleSense generates a grounded Arabic response combining detected outfit context with retrieved fashion knowledge.
9. Source References
10. The system displays the top 3 retrieved knowledge chunks used to ground the recommendation.





RESULTS & FUTURE WORK

End-to-end pipeline works | FastAPI + Streamlit integration | Automated tests
Future: Color detection • Clothing attributes • Larger knowledge base • Personalization • Deployment

